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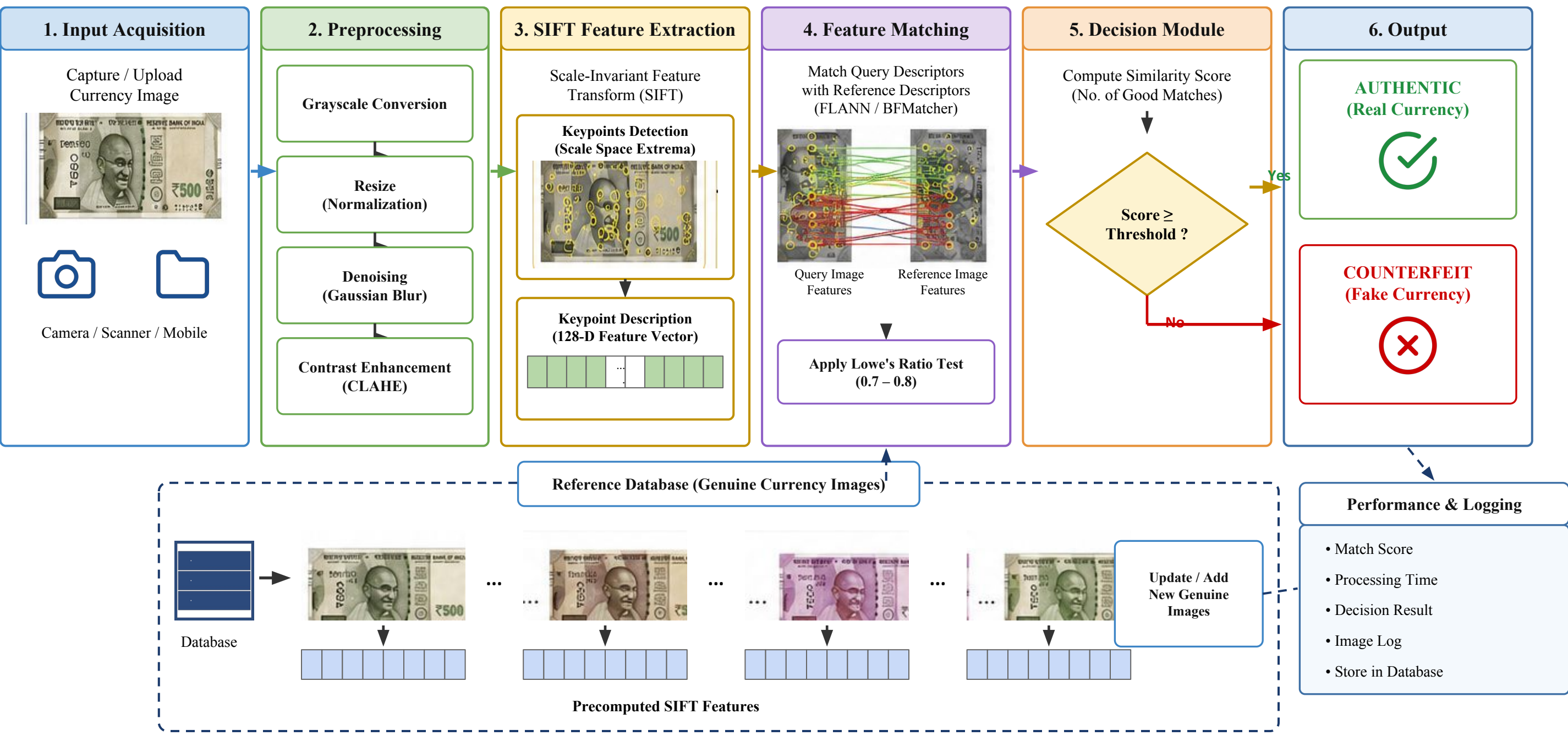
SIFT Feature Matching for Counterfeit Currency Detection

DSA0203-Computer Vision with Open CV

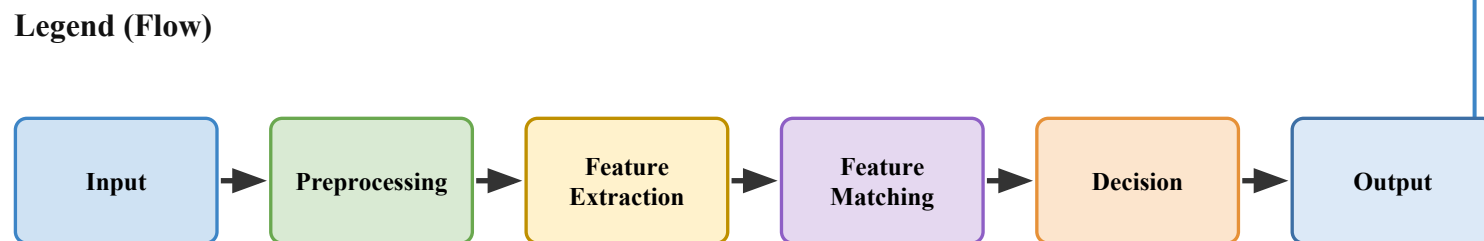
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System Architecture



- Key Features**
- Scale, Rotation & Illumination Invariant
 - Robust to Noise and Viewpoint Changes
 - High Distinctiveness for Currency Patterns
 - Real-time Detection Capability





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Abstract

Counterfeit currency detection is an important application of image processing and computer vision for identifying fake banknotes accurately and efficiently. This system uses Scale-Invariant Feature Transform (SIFT) to detect and describe distinctive features present on a currency note. The input currency image is first pre processed by converting it to grayscale and improving its quality. SIFT is then used to extract key points and feature descriptors from the input image. These features are compared with the features of a genuine currency reference image using feature matching techniques such as BFMatcher or FLANN. The number of reliable feature matches is analyzed using a similarity threshold to determine whether the currency is authentic or counterfeit. Since SIFT features are relatively robust to changes in scale, rotation, and illumination, the proposed approach can provide reliable matching under different image conditions. The system demonstrates how feature-based image processing can be applied to automated currency verification.